

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	Hard

Question

Neurobiologists Laura Cuaya, Raúl Hernández-Pérez, and colleagues investigated the language detection abilities of eighteen dogs. The researchers monitored the brain activity of Joey (an Australian shepherd), Mini (a mixed breed), and other dogs while the animals listened to three recordings: one of *The Little Prince* being read in Spanish, the second in Hungarian, and a third made up of short, randomly selected fragments of the first two, scrambled so that they didn't resemble human speech. Each dog was familiar with either Spanish or Hungarian, but not both. The team concluded that differences in dogs' anatomical features may affect their ability to distinguish speech from nonspeech.

Which finding from the study, if true, would most directly support the team's conclusion?

Answer

- A. Long-headed dogs accustomed to hearing Spanish tended to show more brain activity when hearing Spanish than long-headed dogs accustomed to hearing Hungarian showed when hearing Hungarian.
- B. Compared with shorter-headed dogs, longer-headed dogs showed a greater difference in brain activity when hearing either Spanish or Hungarian than when hearing the scrambled recording.
- C. The pattern of brain activity that long-headed dogs showed when hearing the scrambled recording was different from the pattern of brain activity that short-headed dogs showed when hearing the language they were accustomed to.
- D. Compared with shorter-headed dogs, longer-headed dogs showed a greater difference in brain activity when hearing the language they were accustomed to than when hearing the other language.

Correct Answer: B

Rationale

Choice B is the best answer because it presents a finding that, if true, would most directly support the research team's conclusion about anatomical features and speech detection in dogs. The text explains that a team of researchers monitored the brain activity of dogs while the dogs listened to three recordings: one of spoken Spanish, one of spoken Hungarian, and one of scrambled fragments that weren't recognizable as human speech. The text then states that the team concluded that differences in dogs' anatomical features may affect their ability to distinguish speech from nonspeech. The finding that longer-headed dogs exhibited a greater difference in brain activity when listening to the speech recordings (in Spanish or Hungarian) versus the nonspeech (scrambled) recording compared with shorter-headed dogs would establish an association between an anatomical feature (head length) and responses (as measured by brain activity) to speech versus nonspeech. This observed relationship between head length and brain activity patterns during exposure to speech and nonspeech would support the team's conclusion.

Choice A is incorrect because it describes a finding about dogs' responses (as indicated by brain activity) to hearing their respective familiar languages rather than dogs' ability to distinguish speech from nonspeech, which is what the team's conclusion specifically addresses. Moreover, this finding pertains to only one anatomical type (long-headed dogs), so it wouldn't support the conclusion that anatomical differences may affect dogs' ability to distinguish speech from nonspeech. Choice C is incorrect because it describes a finding involving a comparison between long-headed dogs listening to nonspeech and short-headed dogs listening to speech, which wouldn't provide enough information to support the researchers' conclusion that anatomical differences may affect dogs' ability to distinguish speech from nonspeech. To support that conclusion, the finding would need to show how dogs with different anatomical features (for example, long-headed and short-headed dogs) respond to recordings of speech as well as recordings of nonspeech. Choice D is incorrect because it describes a finding about dogs' ability to distinguish between a familiar language and an unfamiliar one, not between speech and nonspeech. While this finding does compare dogs with different anatomical features (longer-headed vs. shorter-headed ones), it focuses on language recognition (Spanish vs. Hungarian) rather than the ability to distinguish speech from nonspeech, which is what the team's conclusion specifically addresses.